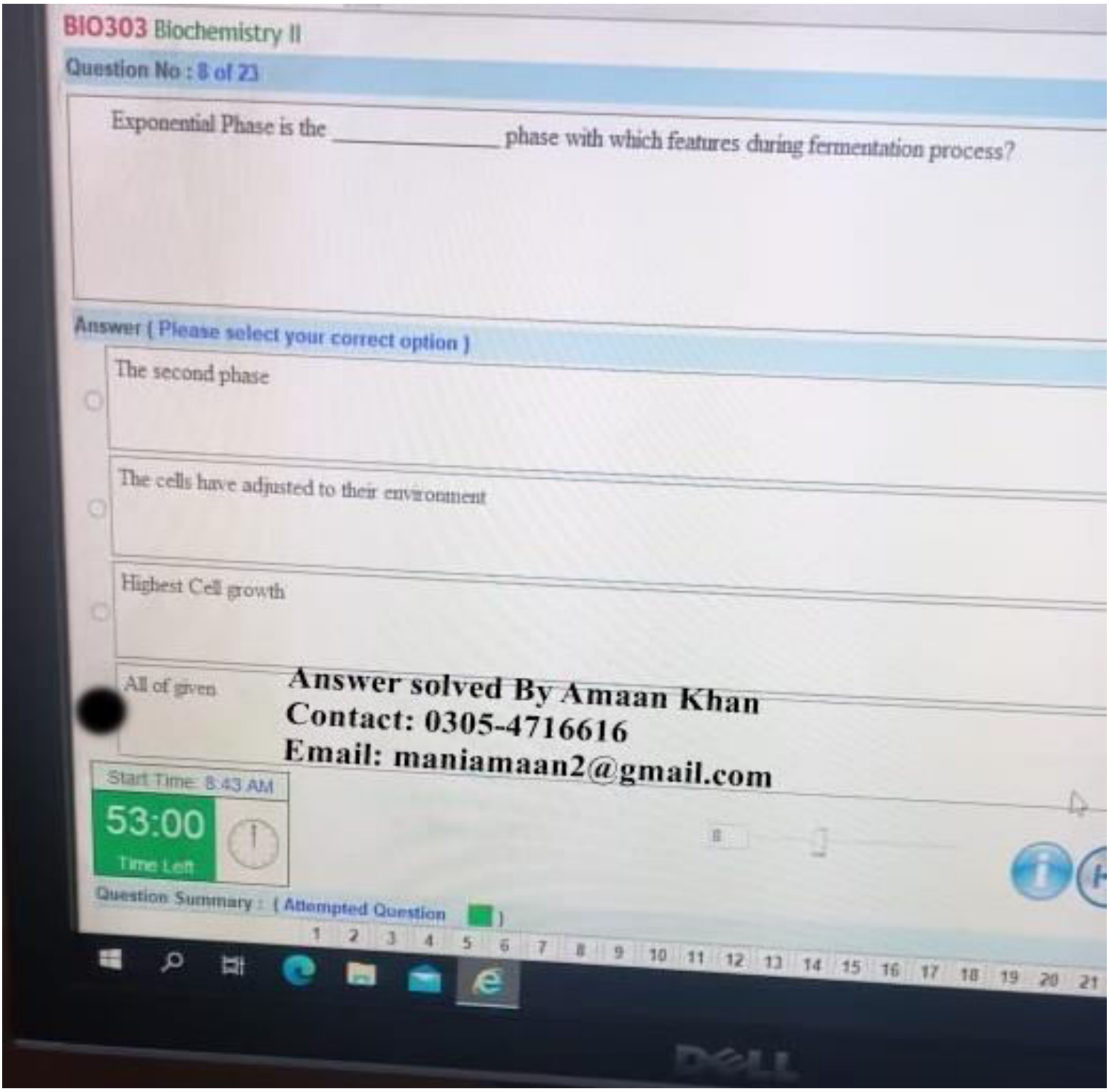
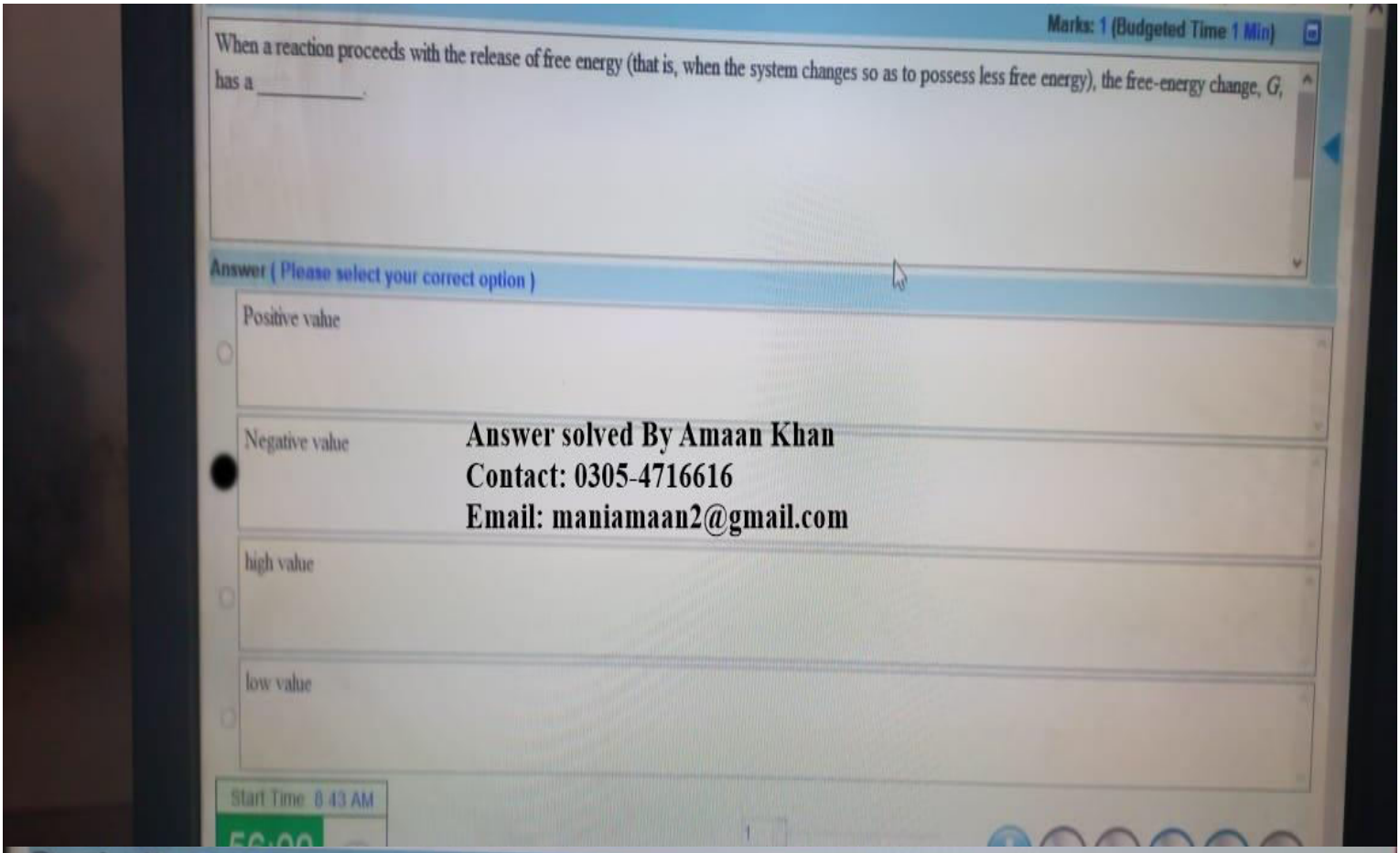
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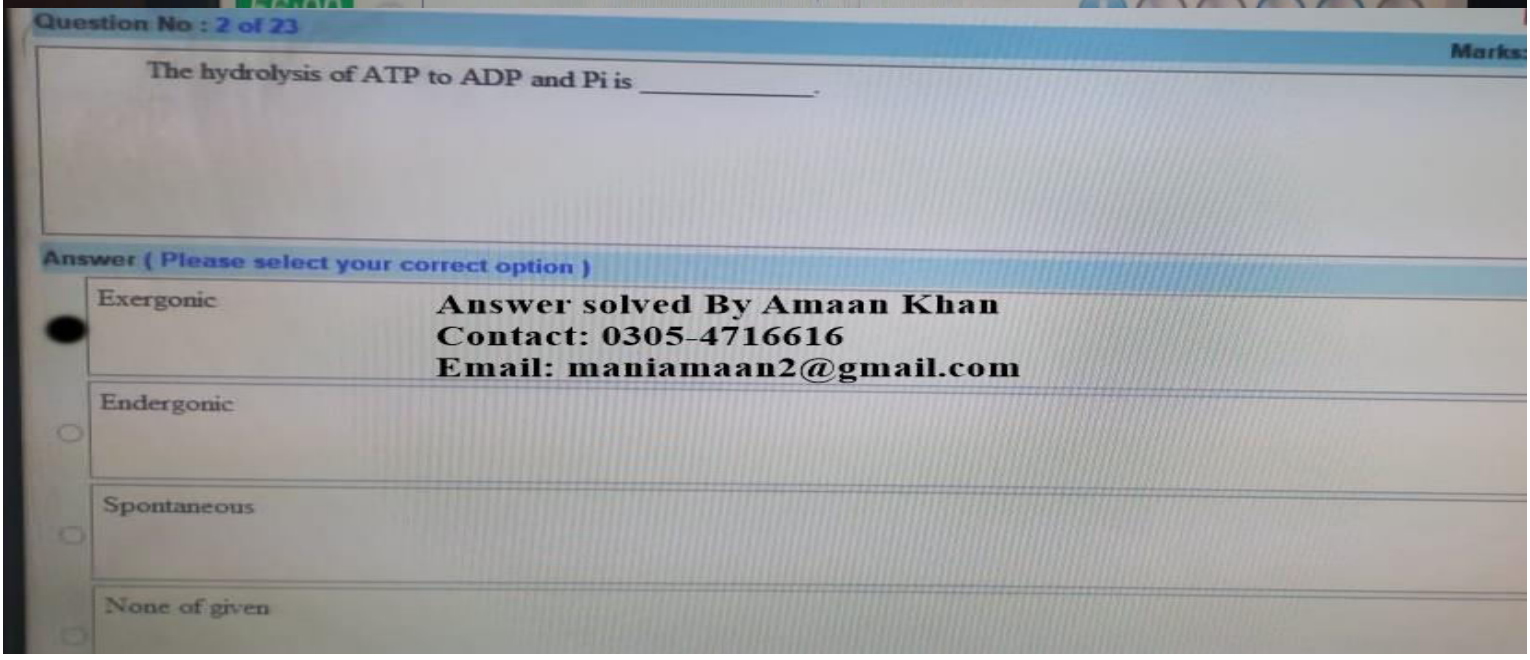
BIO 303 Midterm Fall 2021



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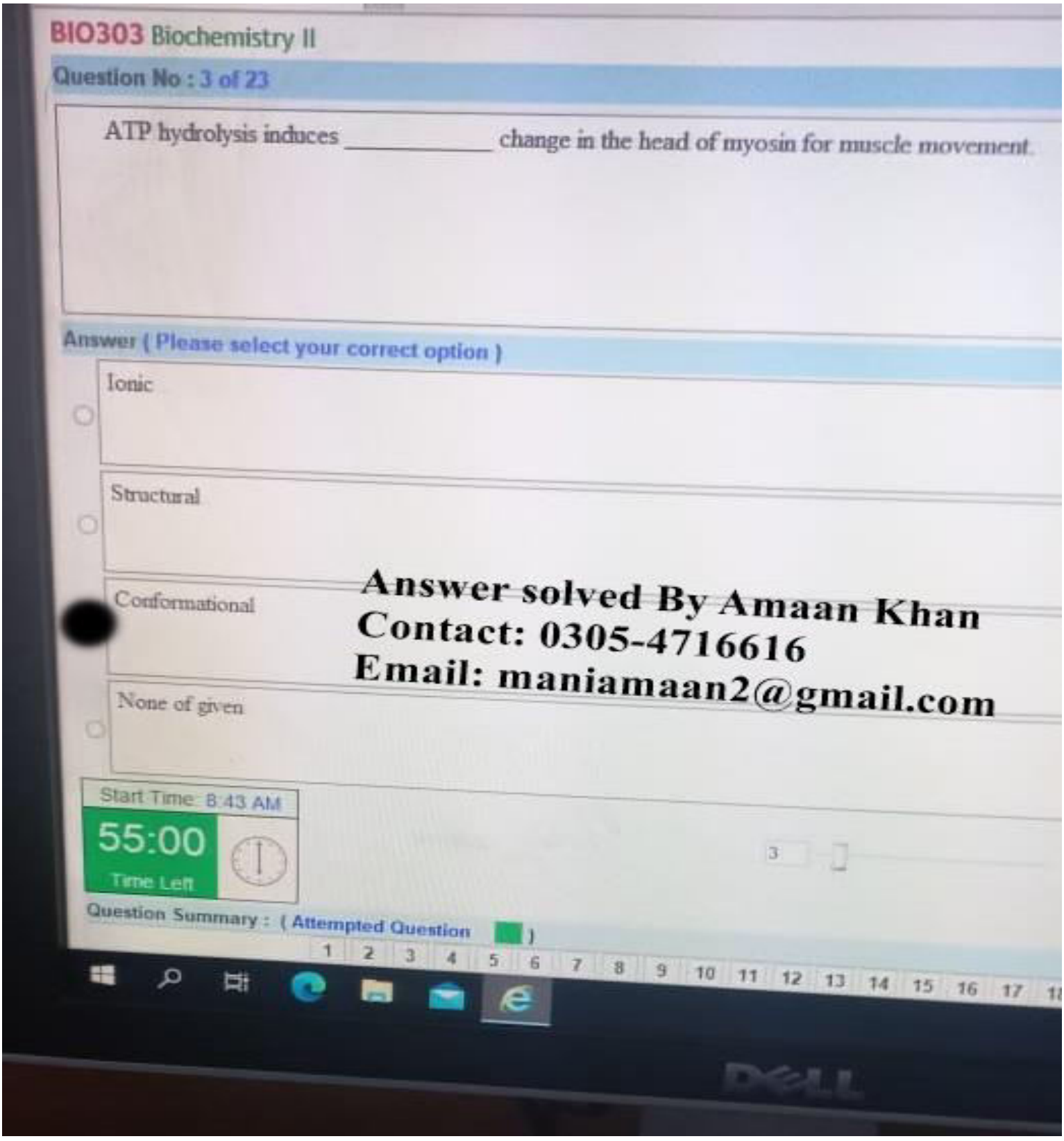
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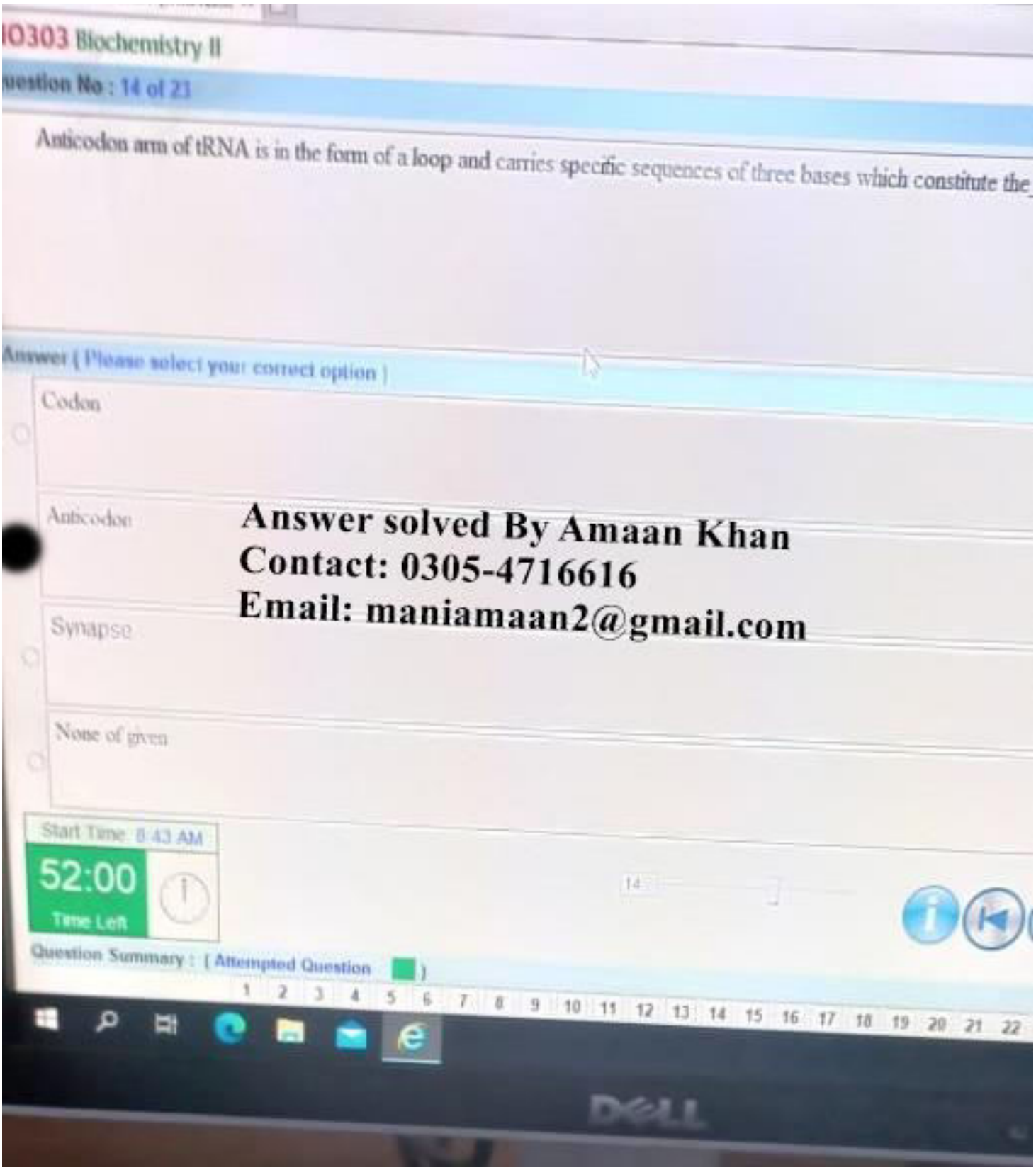
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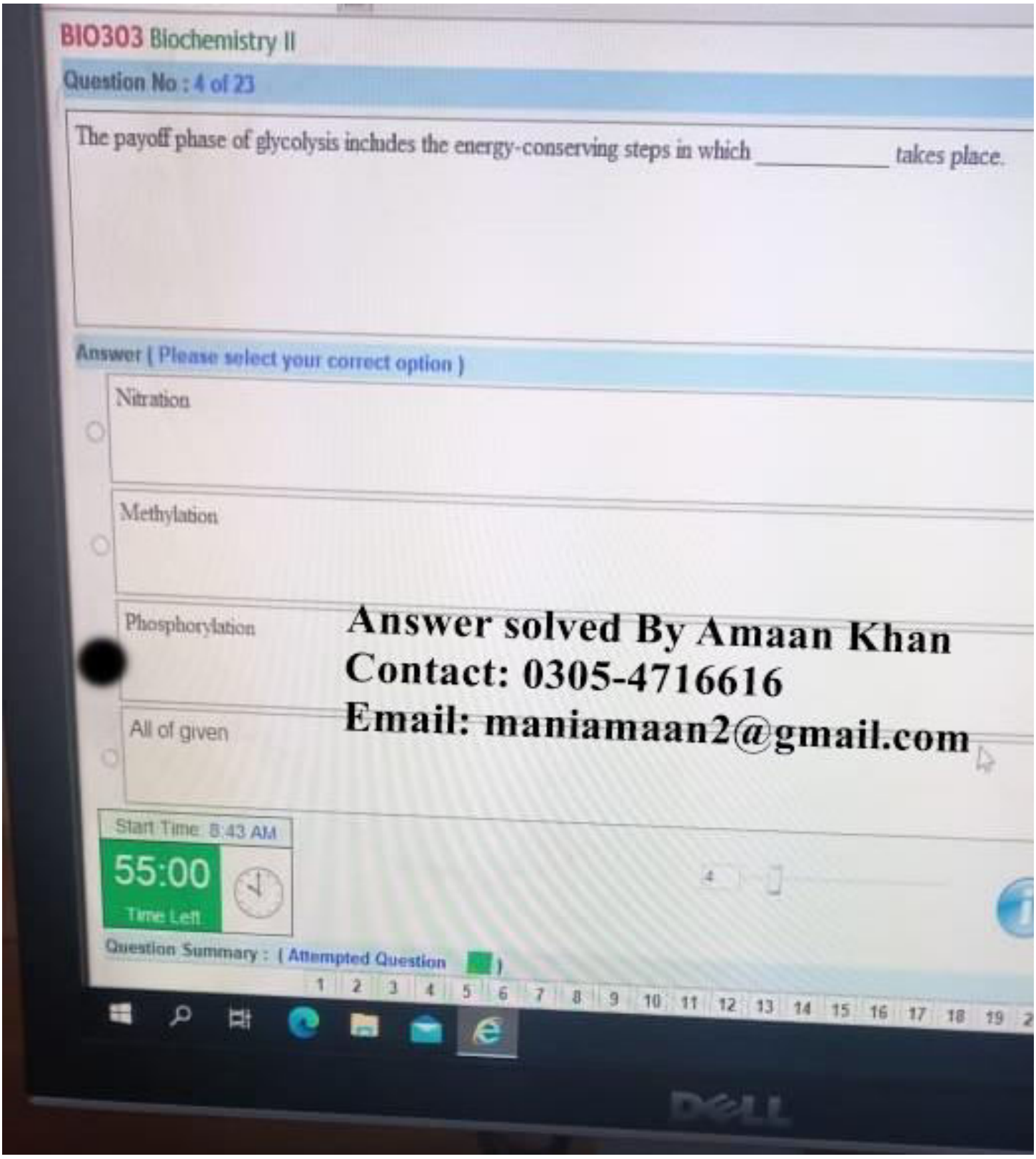
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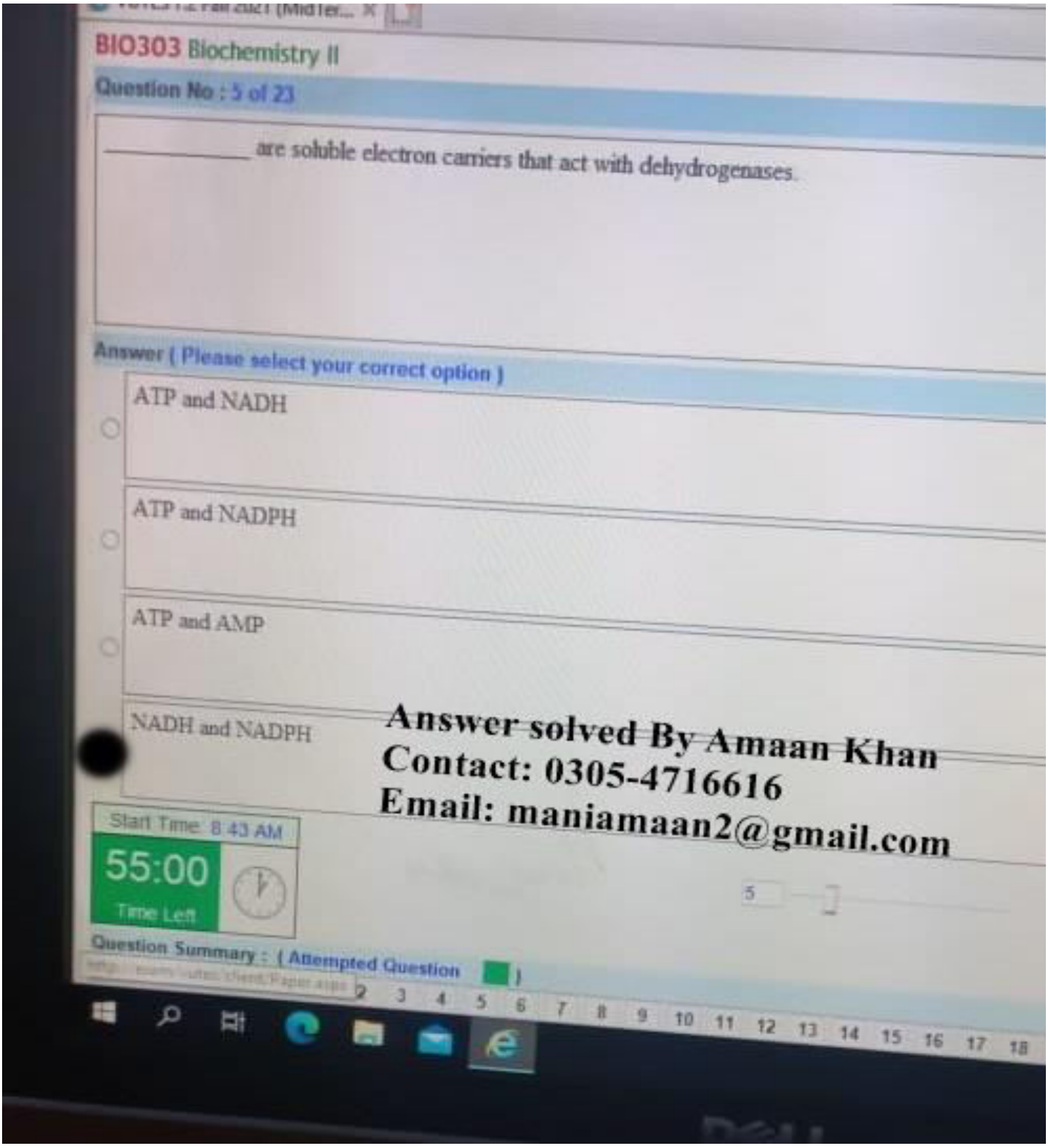
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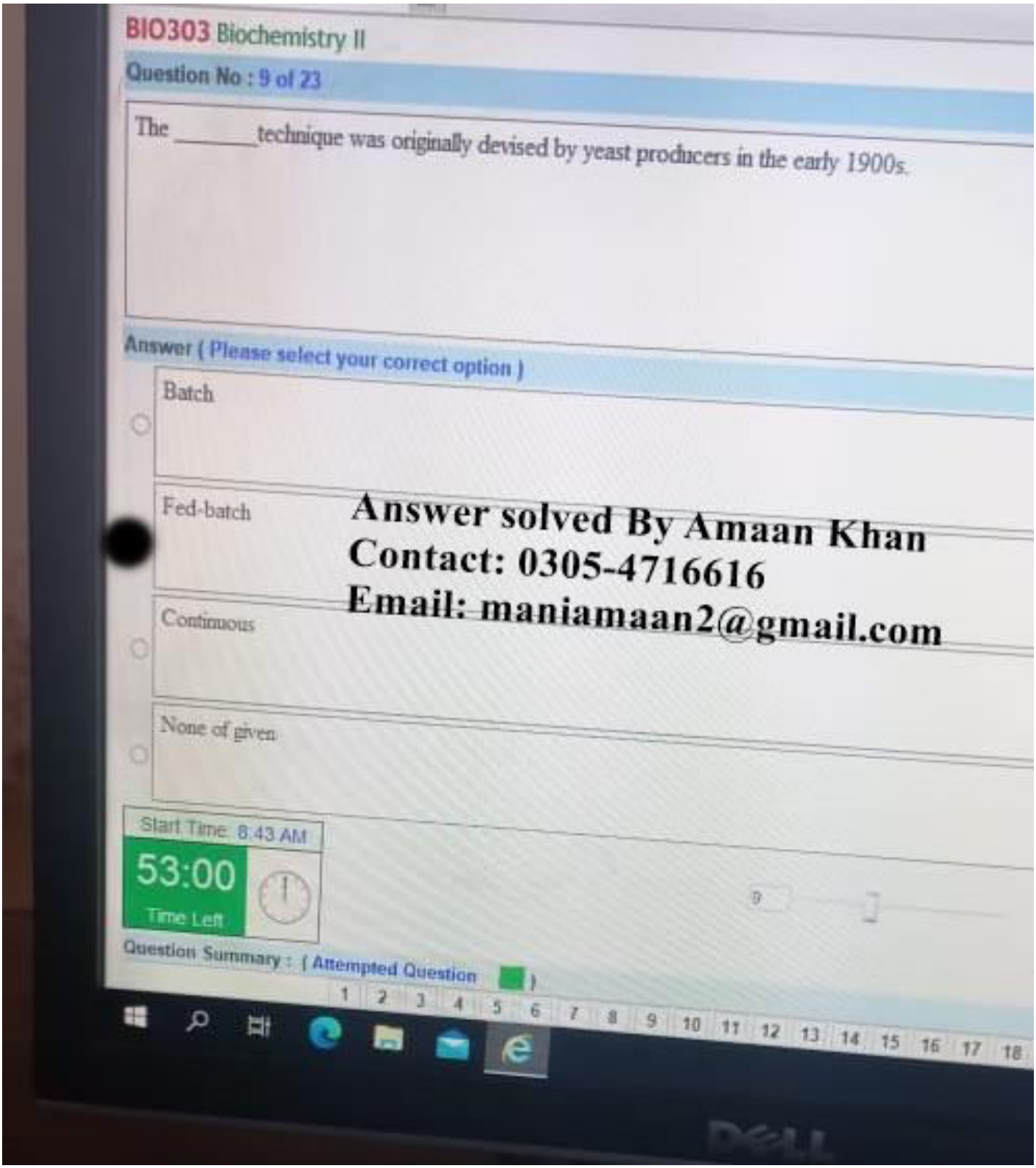
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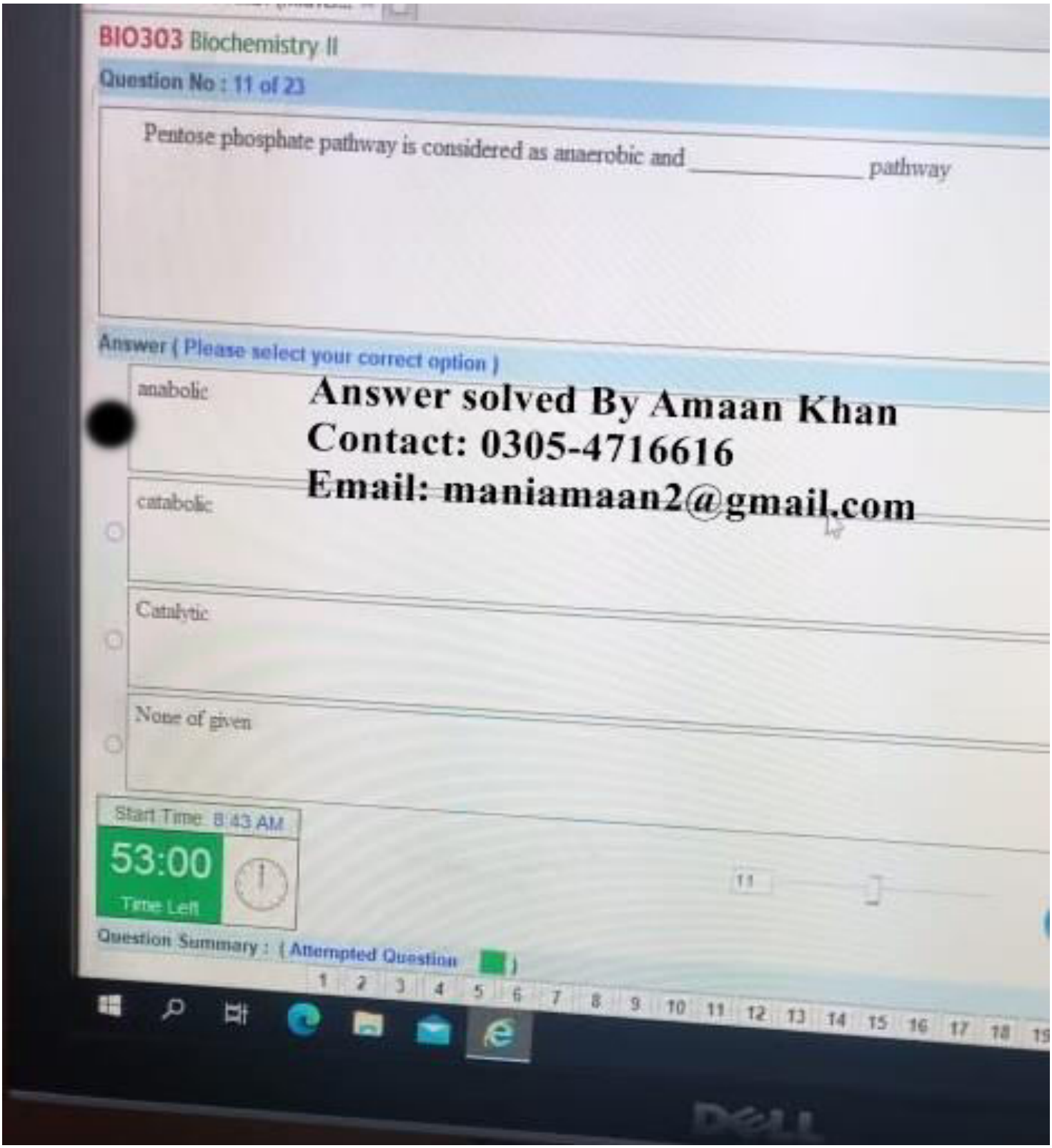
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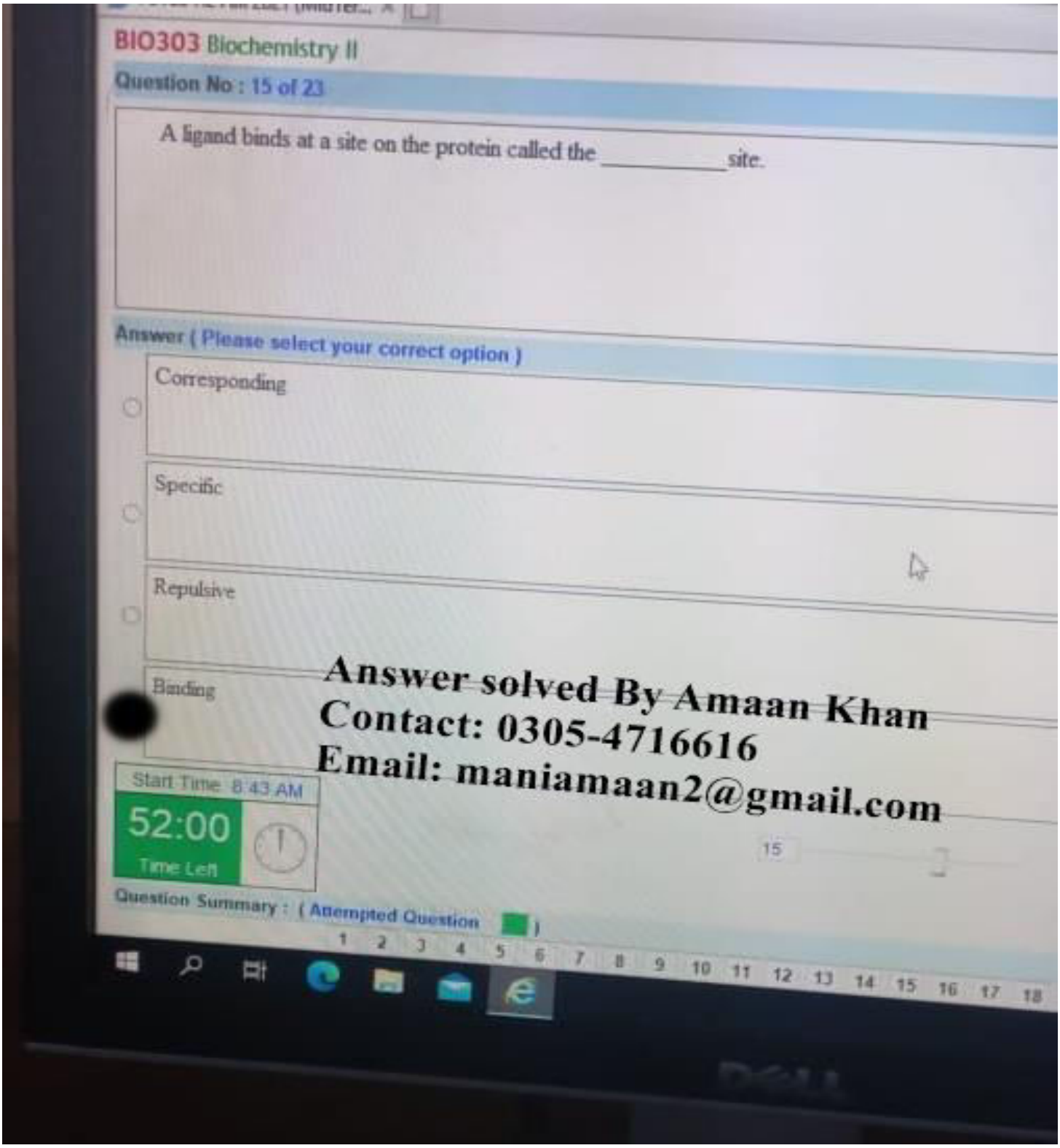
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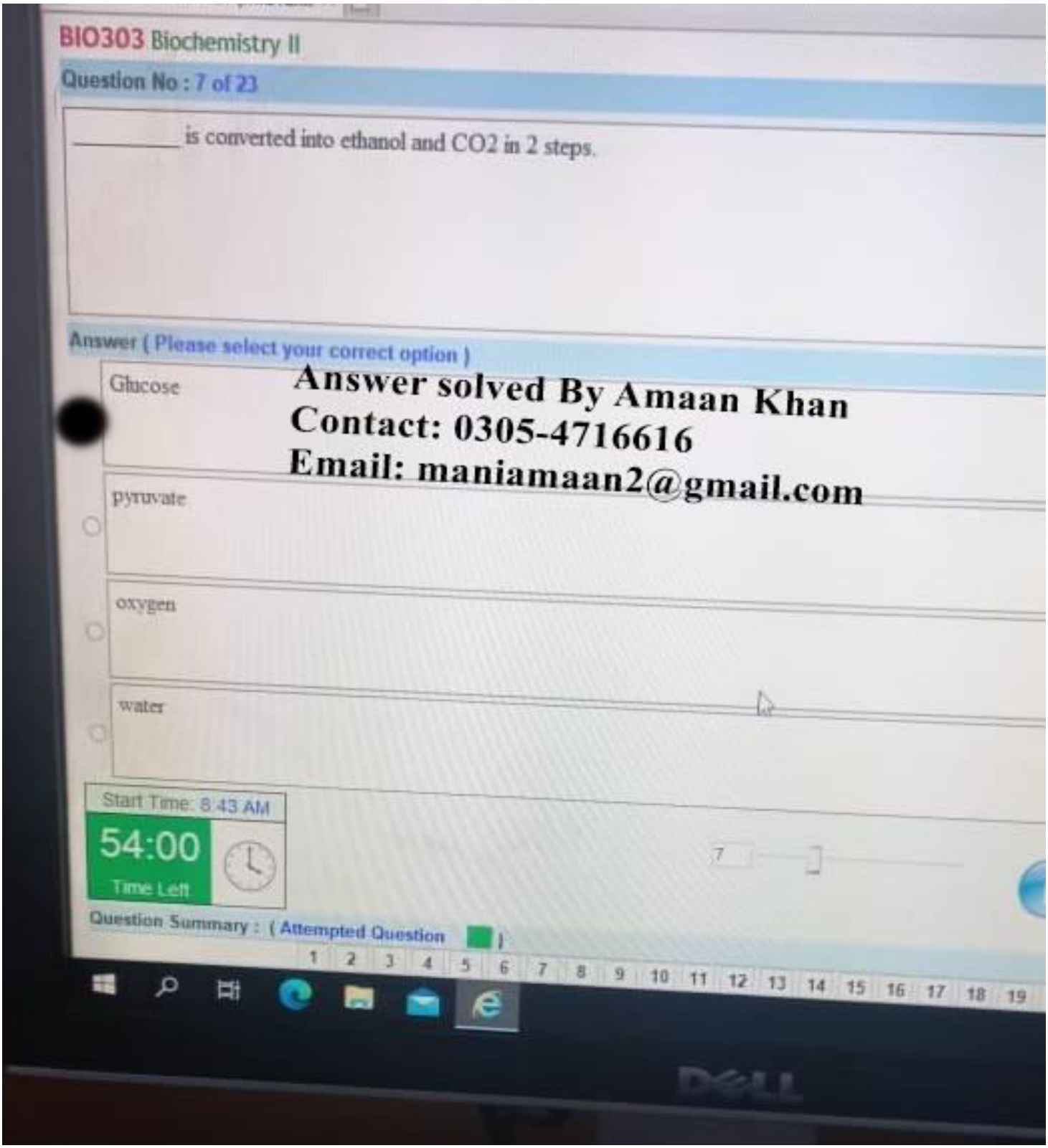
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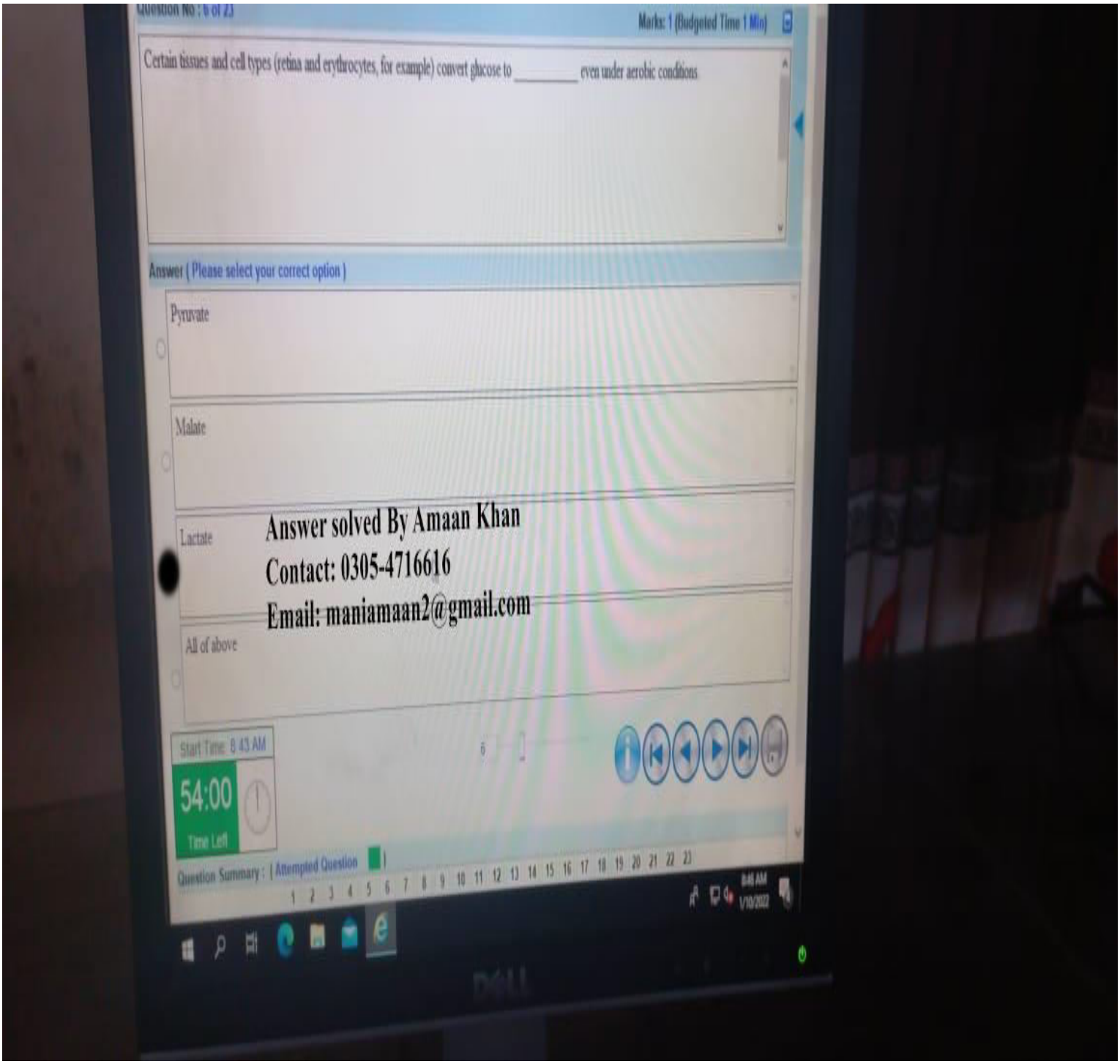
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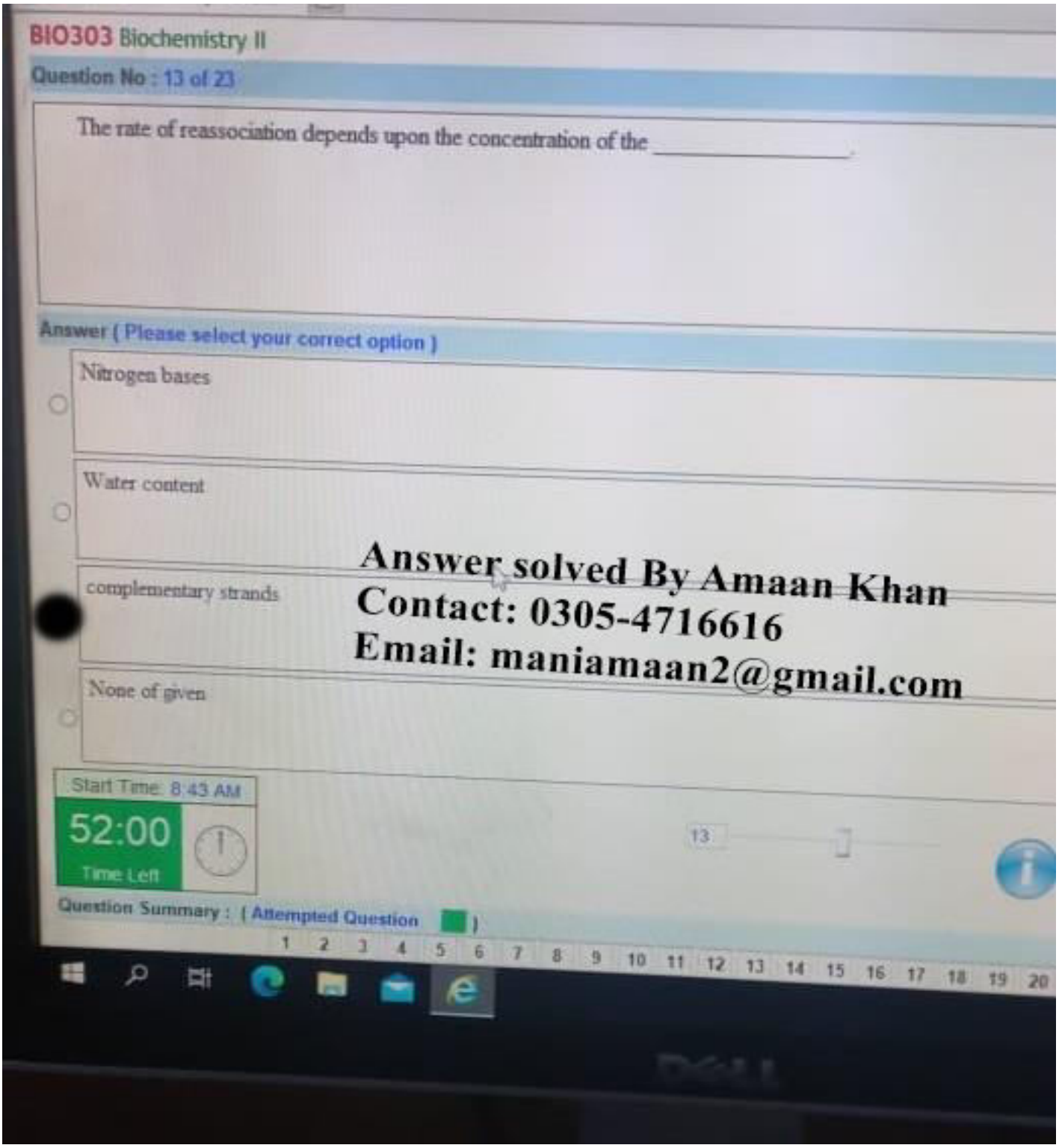
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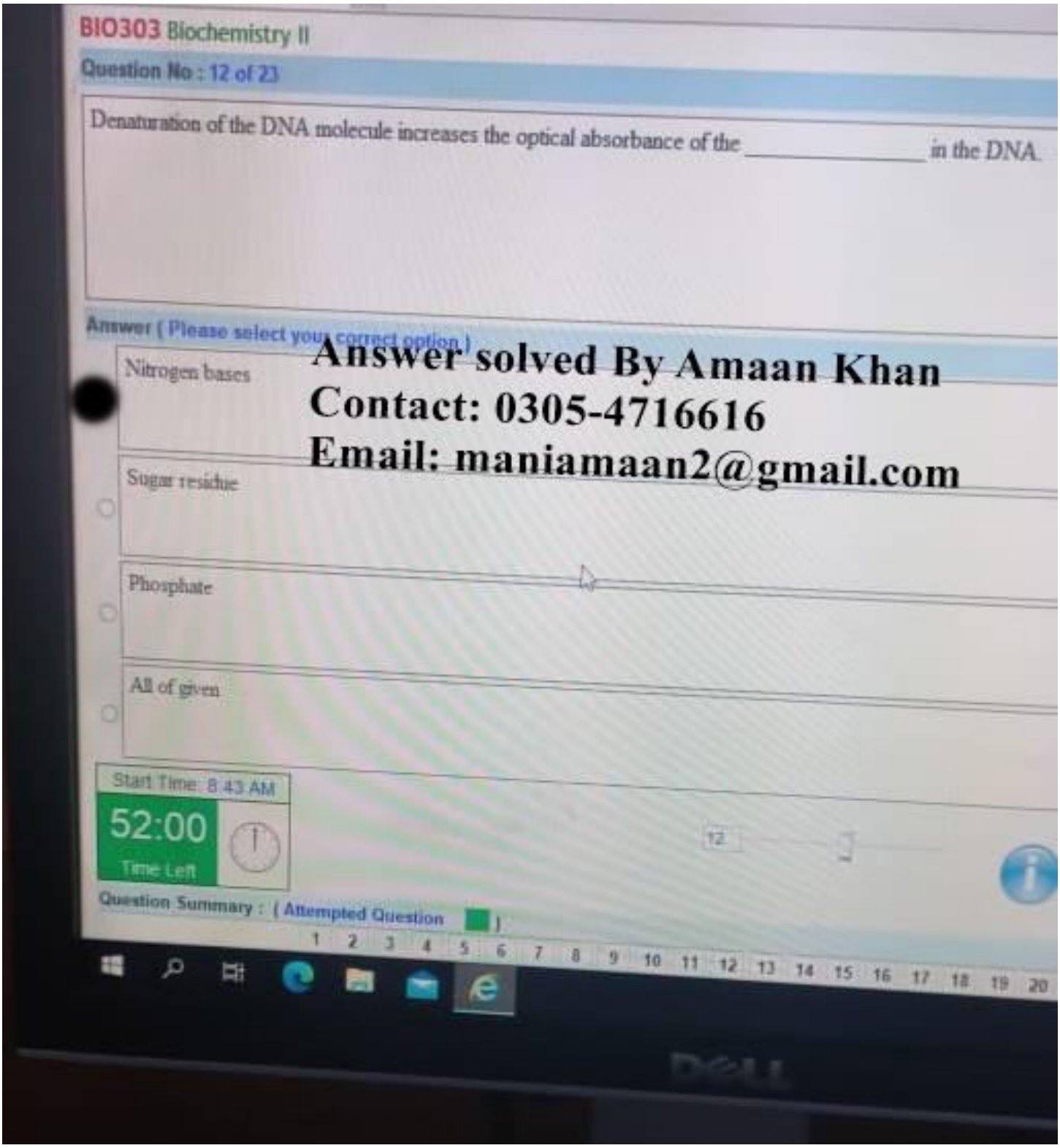
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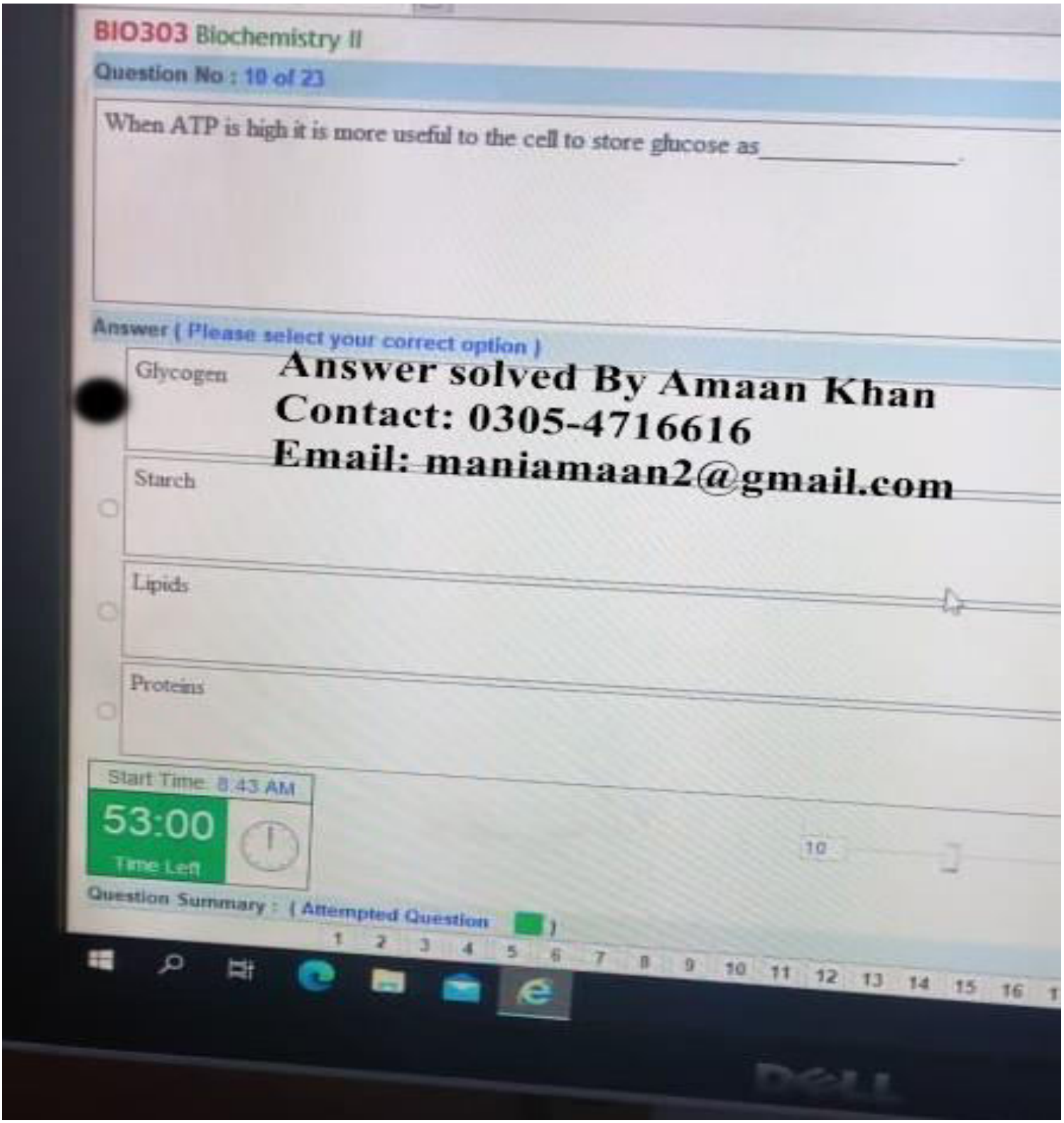
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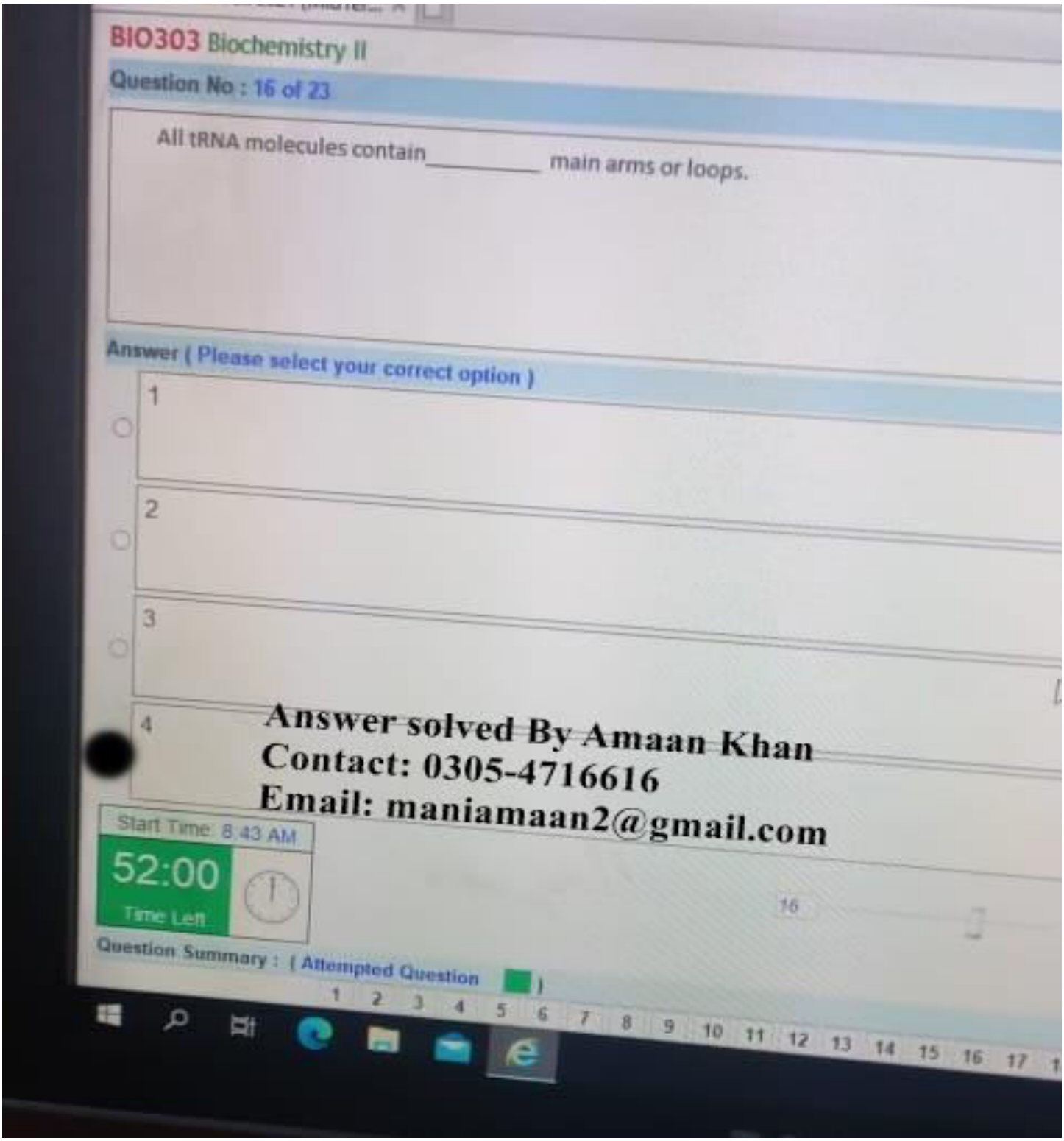
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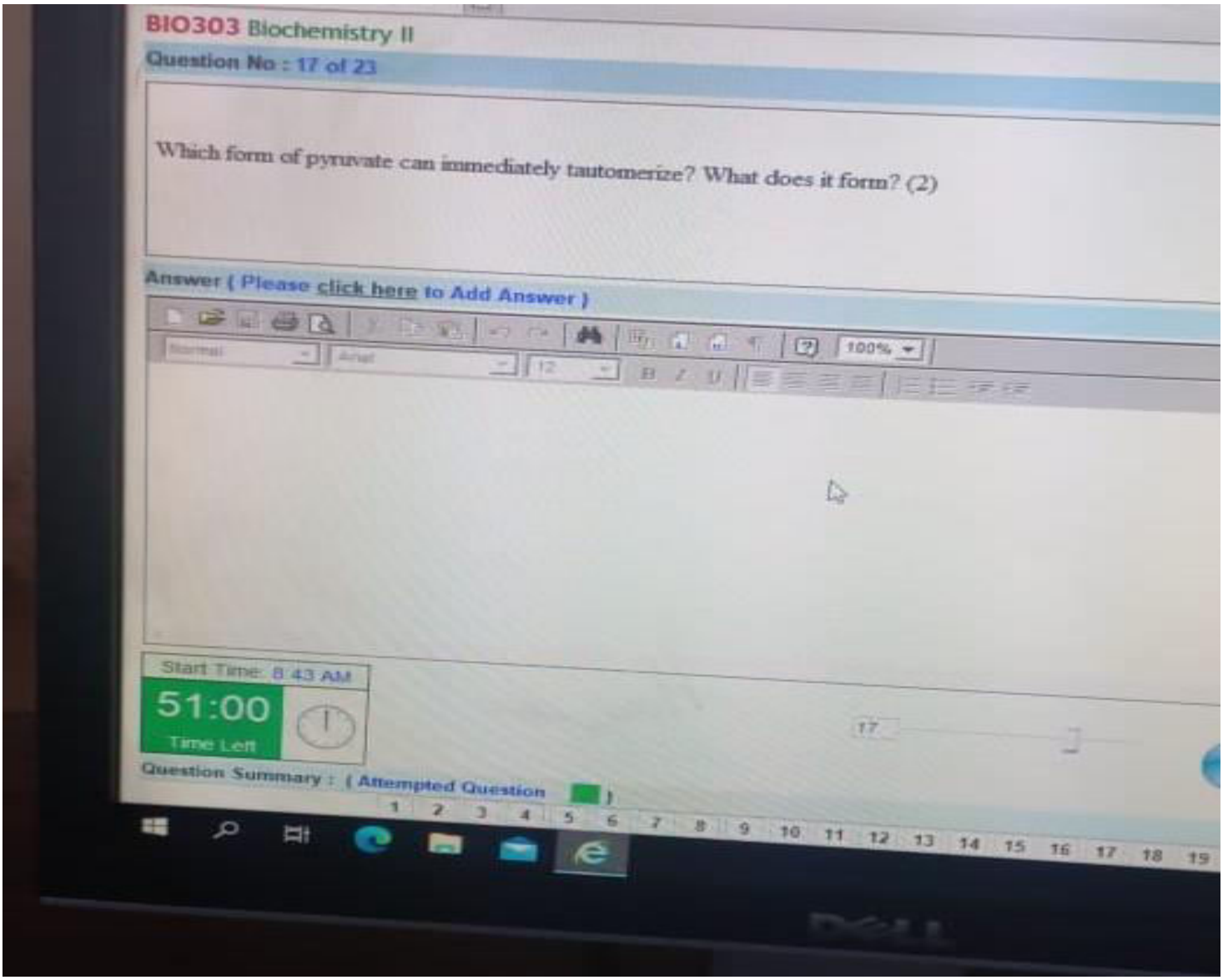
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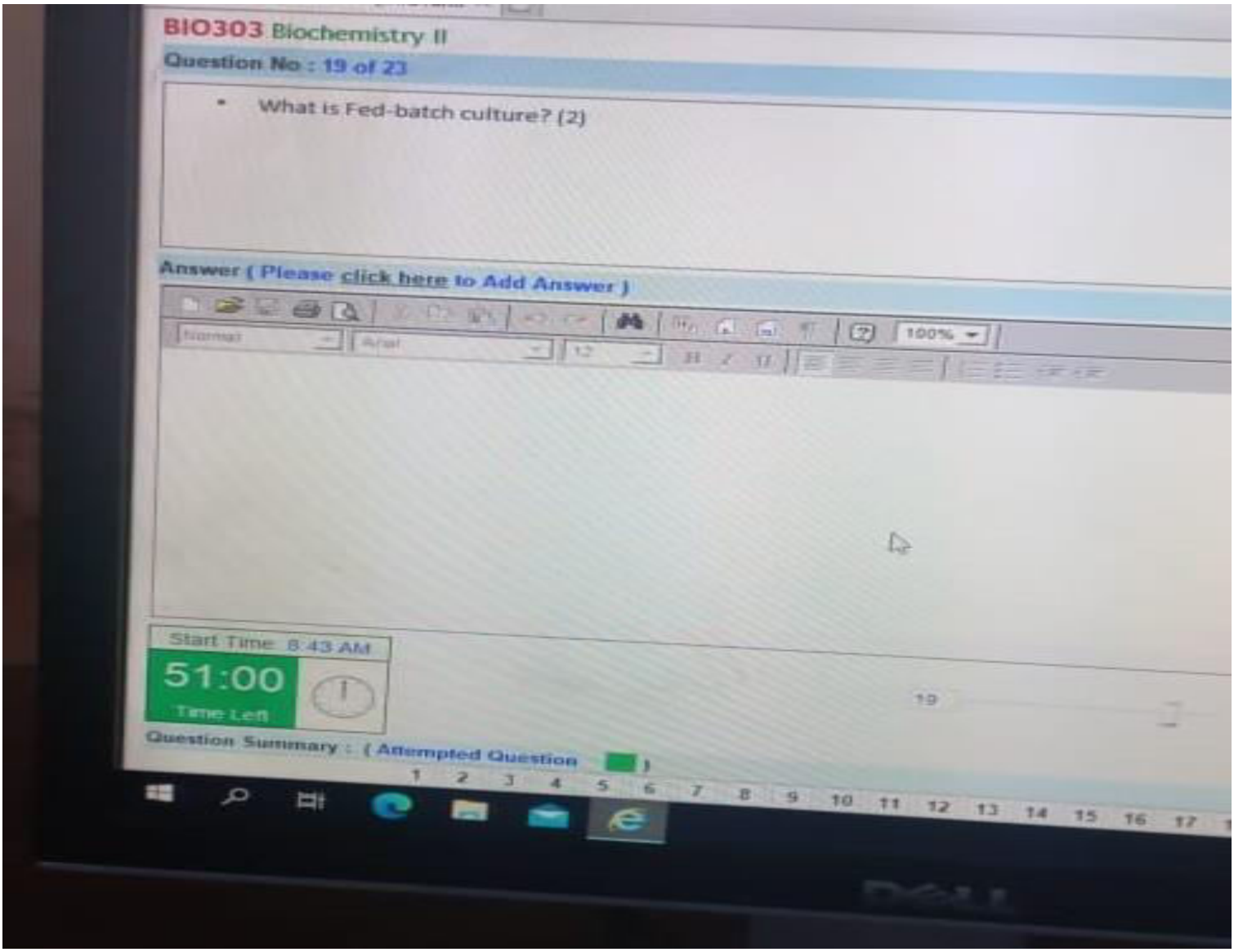


Answer:

In this substrate-level phosphorylation, the product pyruvate first appears in its enol form, then tautomerizes rapidly and nonenzymatically to its keto form, which predominates at pH 7.

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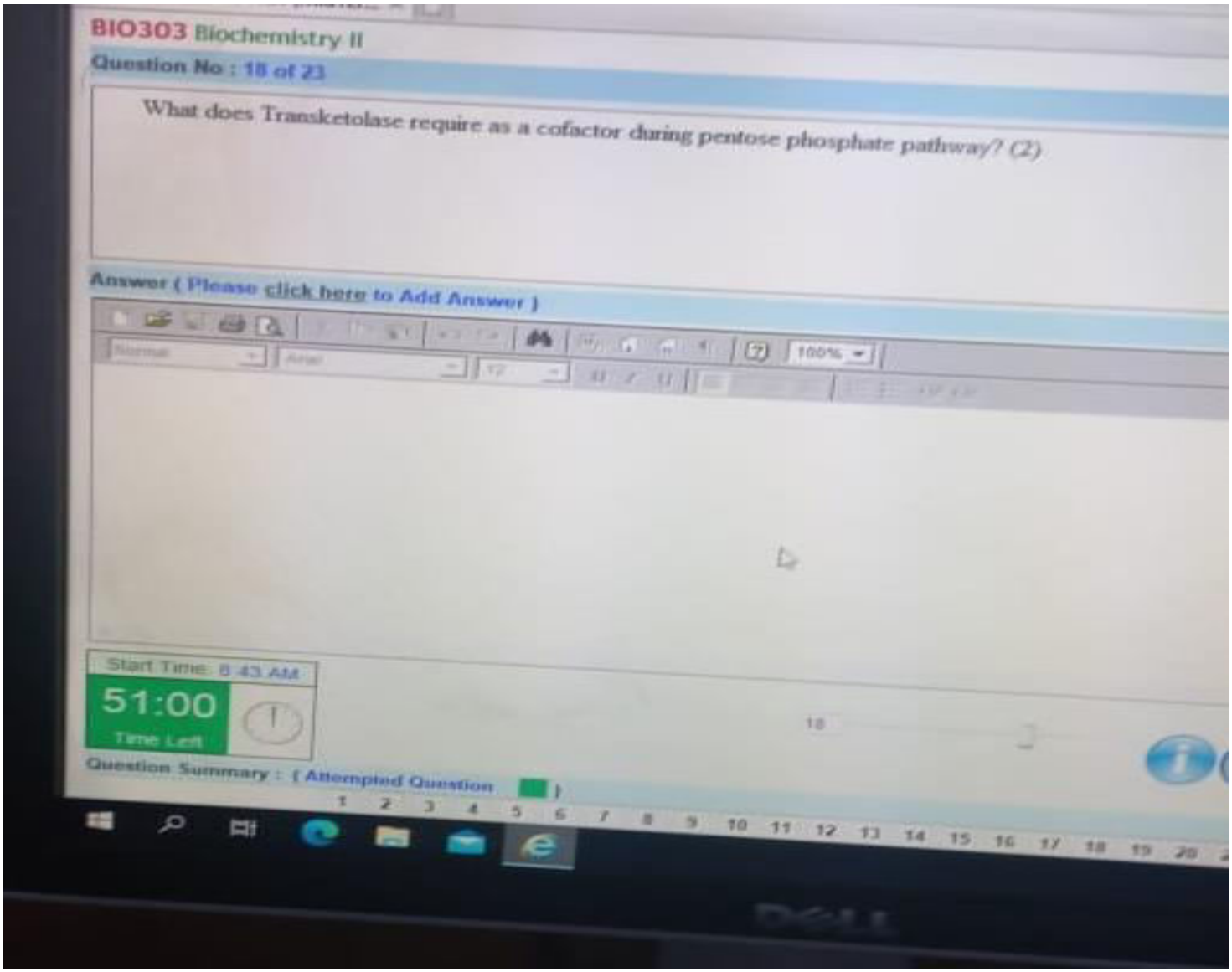


Answer:

Fed-batch culture is, in the broad sense, defined as an operational technique in biotechnological processes where one or more nutrients (substrates) are fed (supplied) to the bioreactor during cultivation and in which the product(s) remain in the bioreactor until the end of the run.

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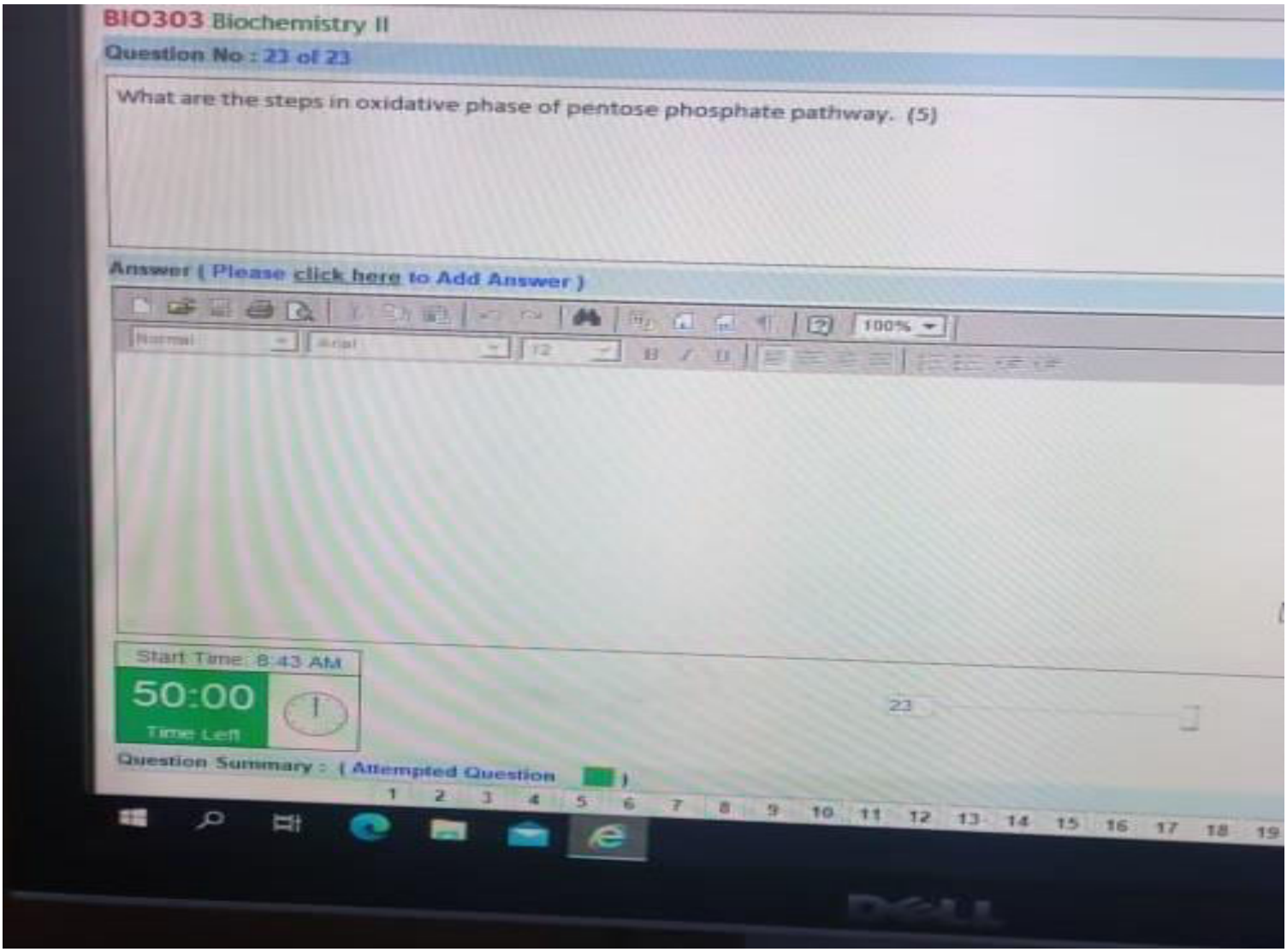


Answer:

Transketolase requires the cofactor thiamine pyrophosphate (TPP), which stabilizes a two-carbon carbanion in this reaction.

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Answer:

1. The Oxidative Phase

1. The Oxidative Phase Produces Pentose Phosphates and NADPH

2. The first reaction of the pentose phosphate pathway is the oxidation of glucose 6-phosphate by glucose 6- phosphate

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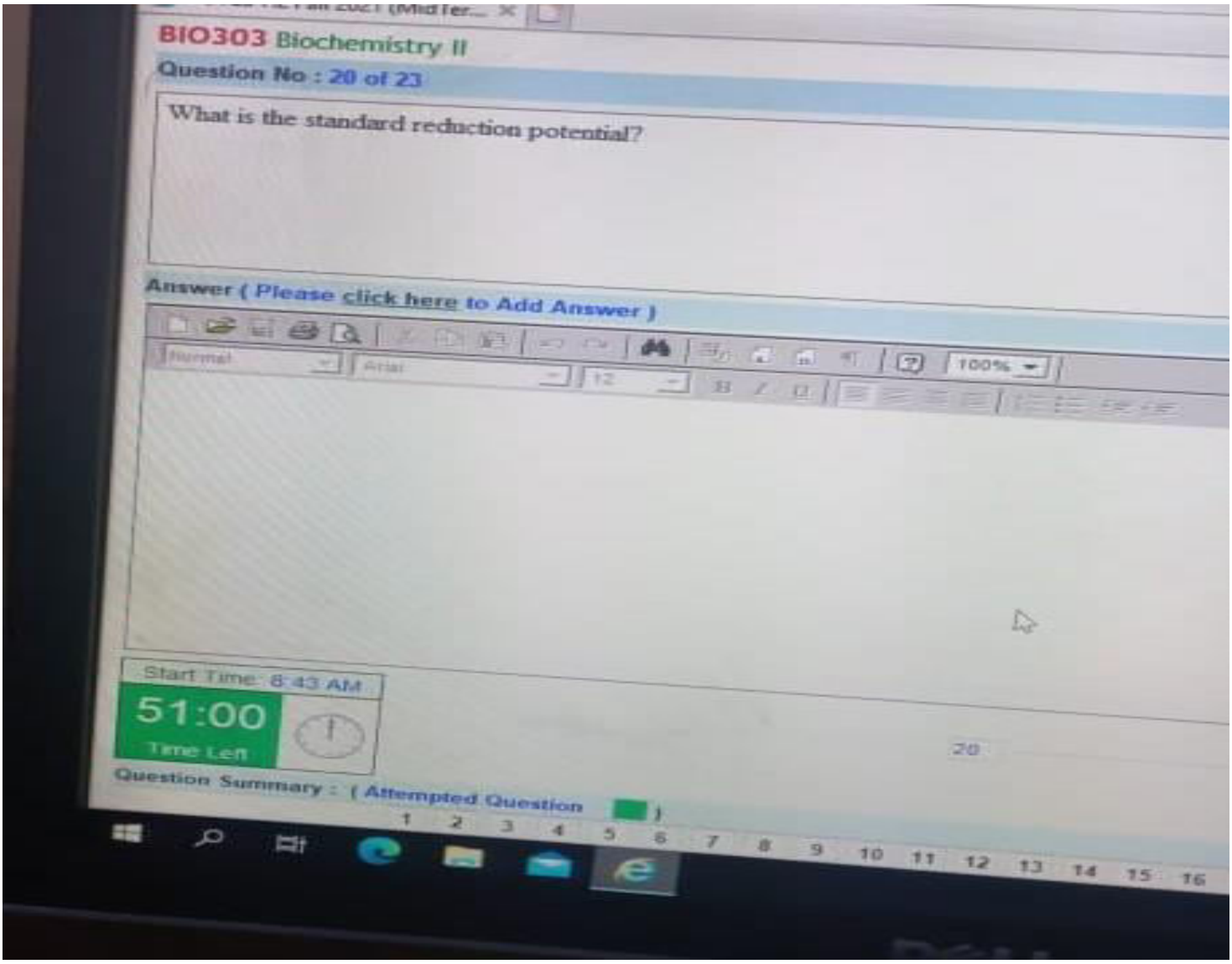
dehydrogenase (G6PD) to form 6- phosphoglucono-delta lactone, an intramolecular

ester.

3. NADP+ is the electron acceptor, and overall equilibrium is in the direction of NADPH formation.

4. The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific enzyme lactonase, then 6-phosphogluconate undergoes oxidation and decarboxylation by 6-phosphogluconate dehydrogenase.

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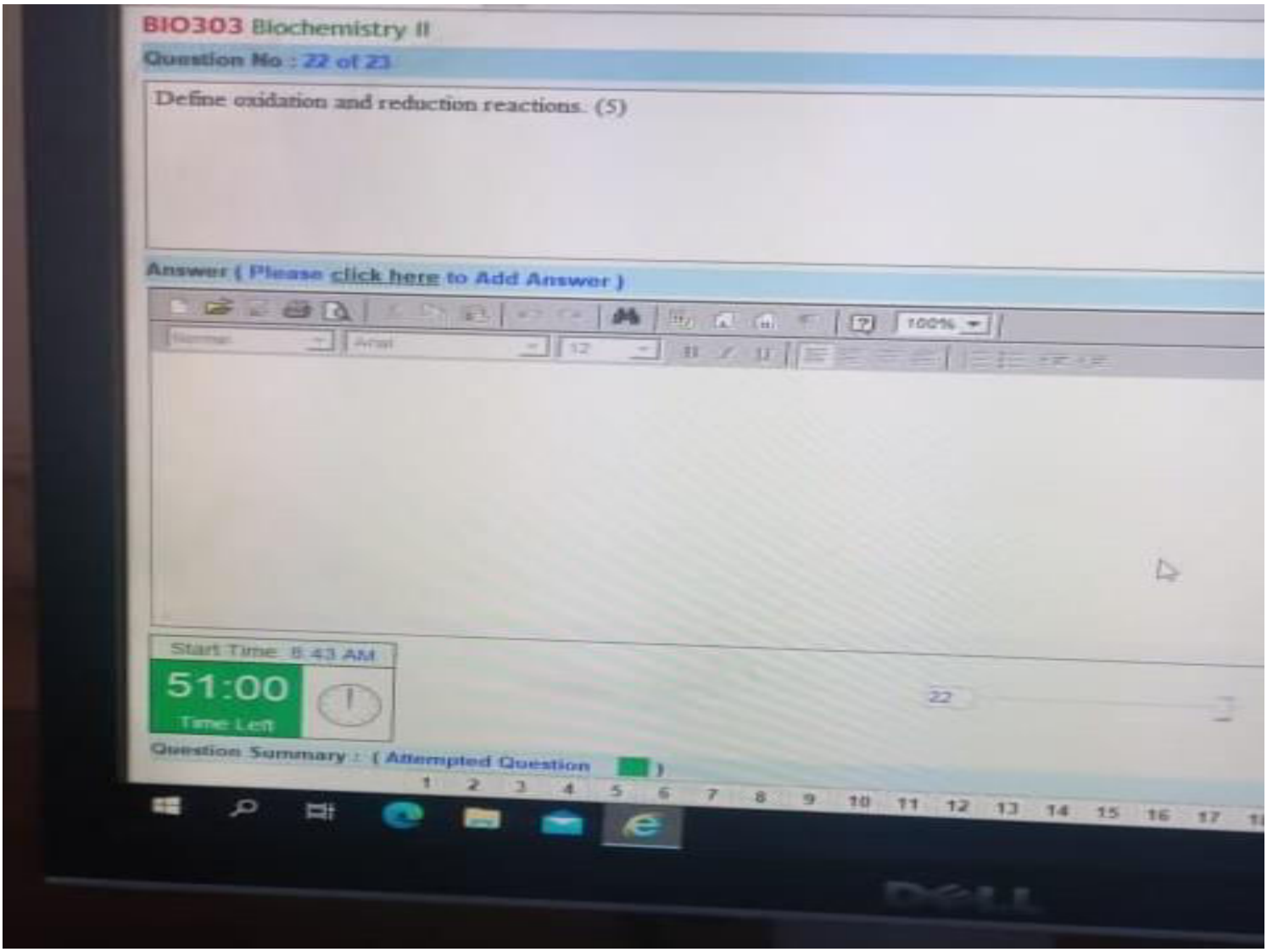


Answer:

The tendency for a chemical species to be reduced, at standard conditions. The more positive the potential is the more likely it will be reduced... denoted by E 0.

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Answer:

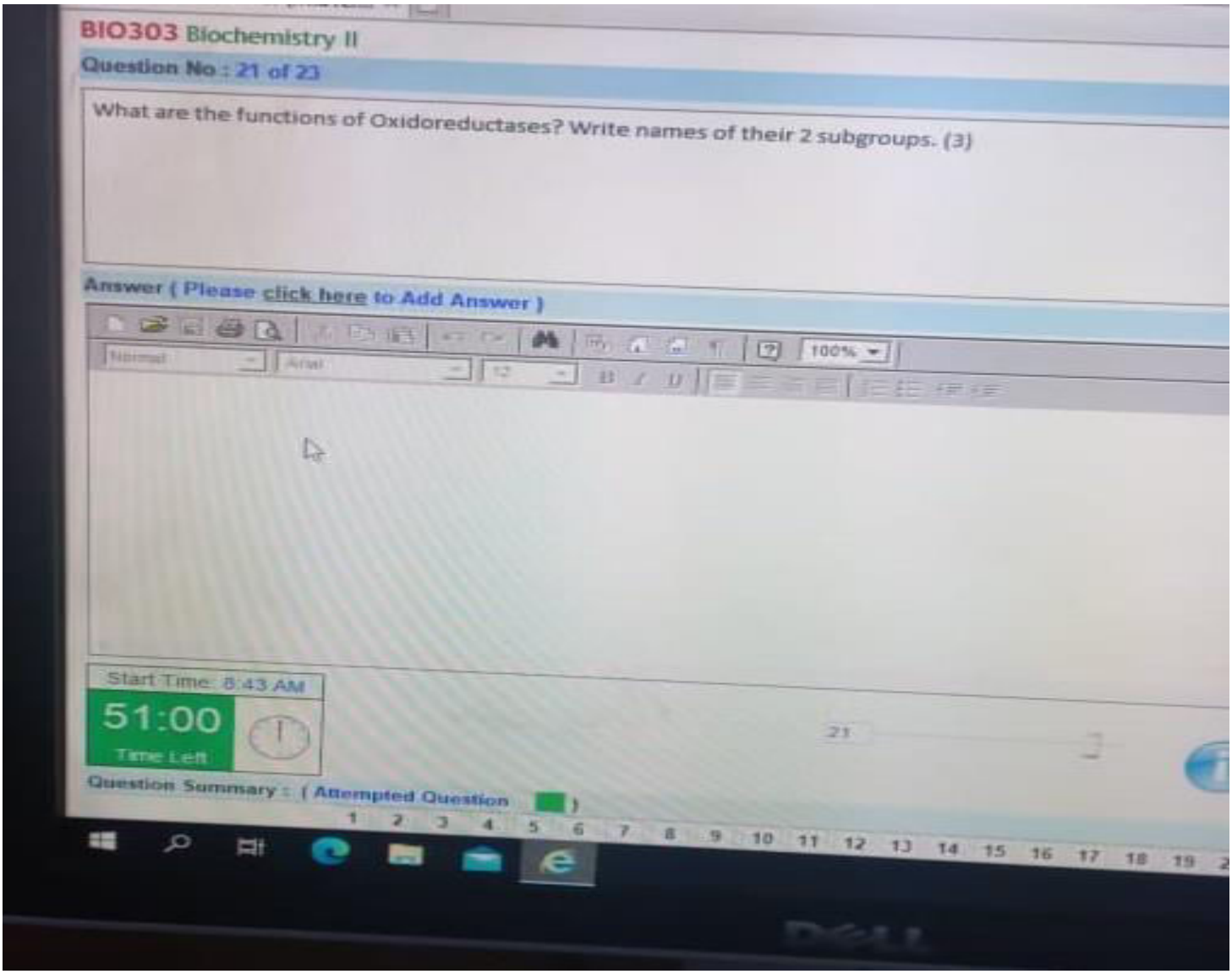
Oxidation

- **Addition of oxygen**
- **loss of electrons**
- **Loss of hydrogen**

Reduction

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- **Removal of oxygen**
- **Gain of electrons**



- **Gain of hydrogen**

Answer:

Oxidoreductases

- **Catalyze oxidation reduction reactions**
- **further divided into four subgroups;**
- **Oxidase,**
- **Dehydrogenases**